VISUAL PAPER SUMMARY

AI makes coding faster. Why doesn't software ship faster?

A visual synthesis of *Artificial Intelligence in the Firm: Bottlenecks in Software Production*

Core finding: local productivity gains are absorbed by downstream review constraints.

Emidio Costa | Executive visual brief

THE EVIDENCE BASE

A large-scale view of real engineering work

WORK EVENTS	FIRMS	WORKERS	OBSERVATION WINDOW
300M	718	725,938	2021-2026

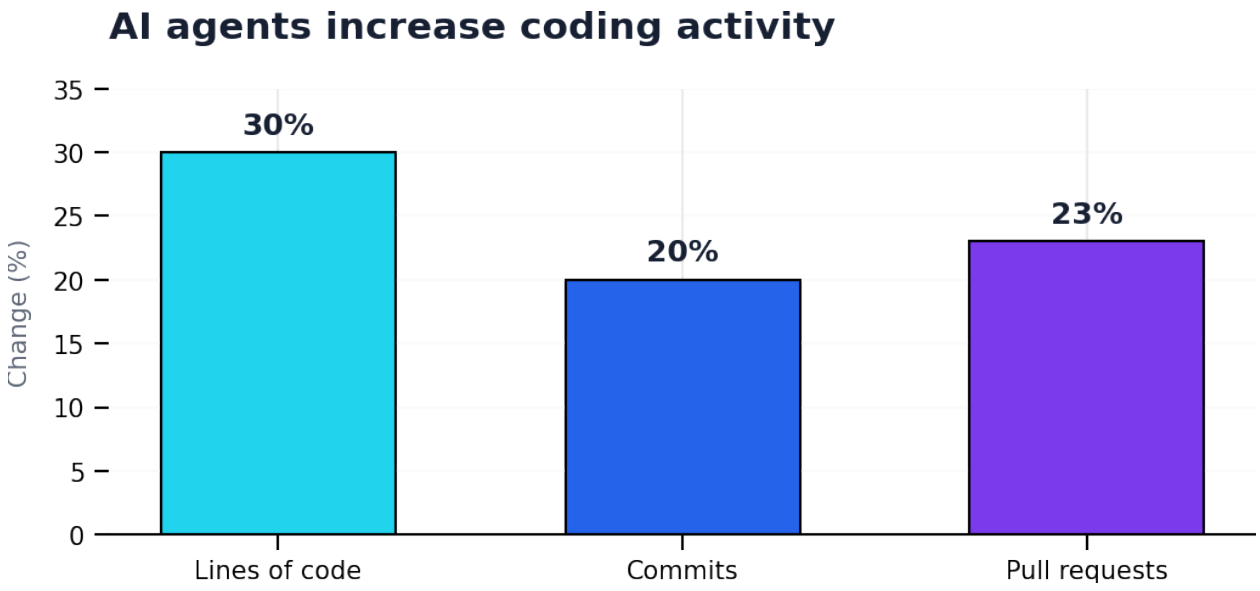
Data connects the full delivery chain



Source: Chen & Stratton, Artificial Intelligence in the Firm: Bottlenecks in Software Production (Aug. 4, 2026)

FINDING 1

AI agents substantially increase coding activity



INTERPRETATION

More draft code

The strongest measured gains appear before review and deployment.

ASSISTANT EFFECT

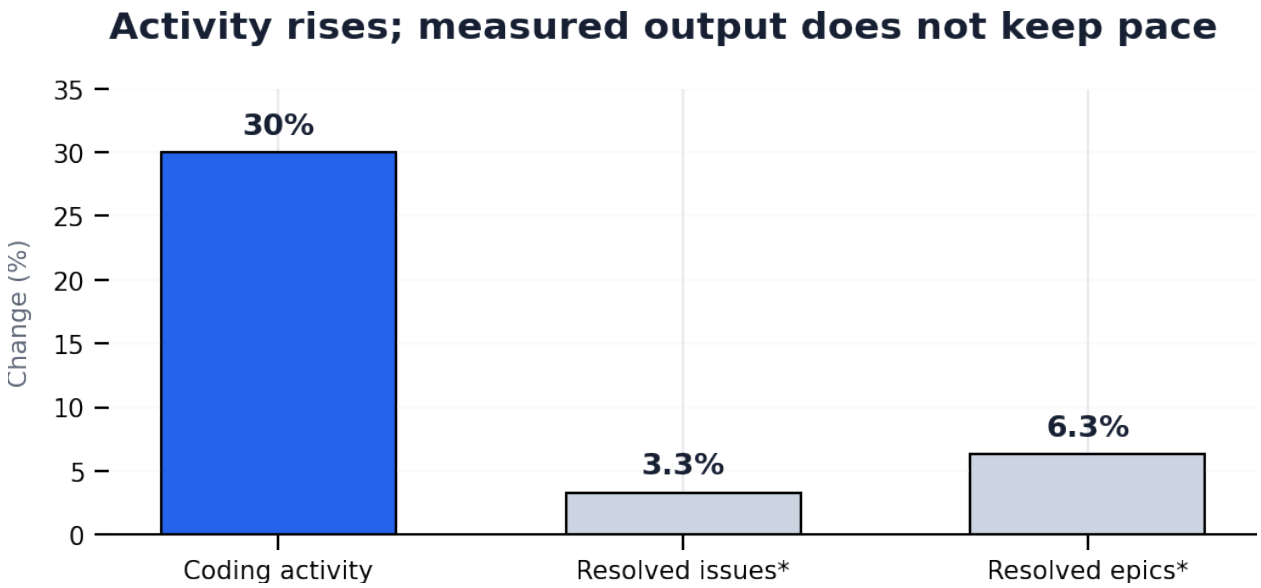
Smaller

Traditional assistants show smaller gains than agents; only commits are statistically significant.

Source: Chen & Stratton, Artificial Intelligence in the Firm: Bottlenecks in Software Production (Aug. 4, 2026)

FINDING 2

More coding does not mean more shipped software



The paper finds no statistically significant increase in resolved Jira issues or epics after AI adoption.

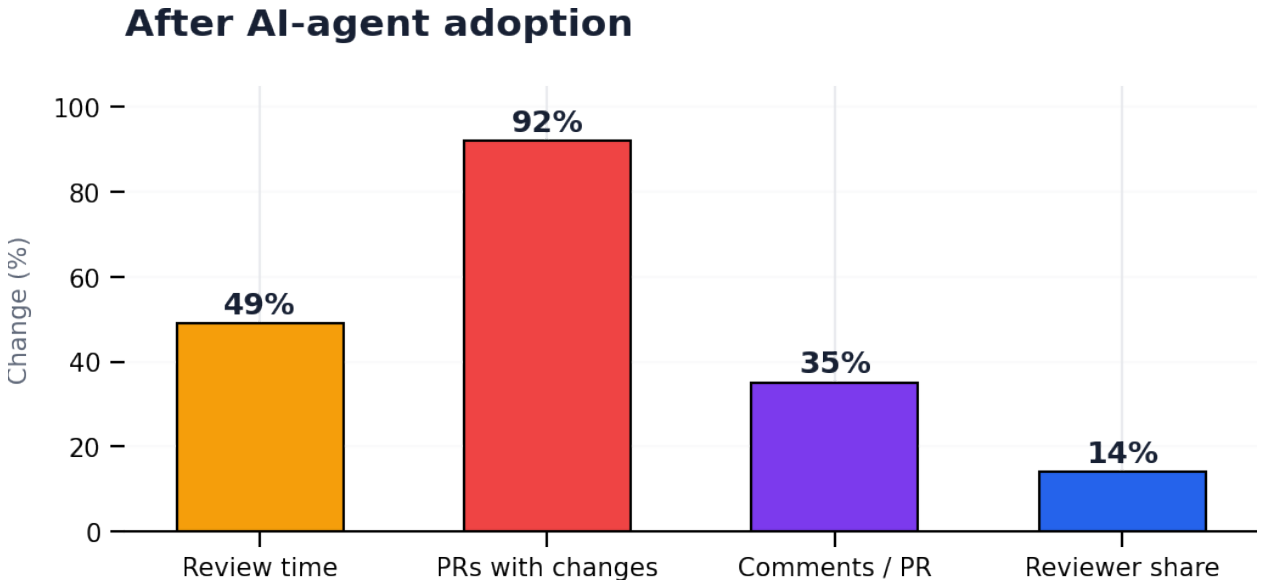
30% more lines of code, but the output effect remains far smaller.

*Visual comparison uses agent point estimates relative to baseline means. The output estimates are not statistically significant.

Source: Chen & Stratton, Artificial Intelligence in the Firm: Bottlenecks in Software Production (Aug. 4, 2026)

FINDING 3

Code review becomes the bottleneck



REVIEW TIME

+49%

Average time from PR submission to merge.

CHANGE REQUESTS

Nearly 2x

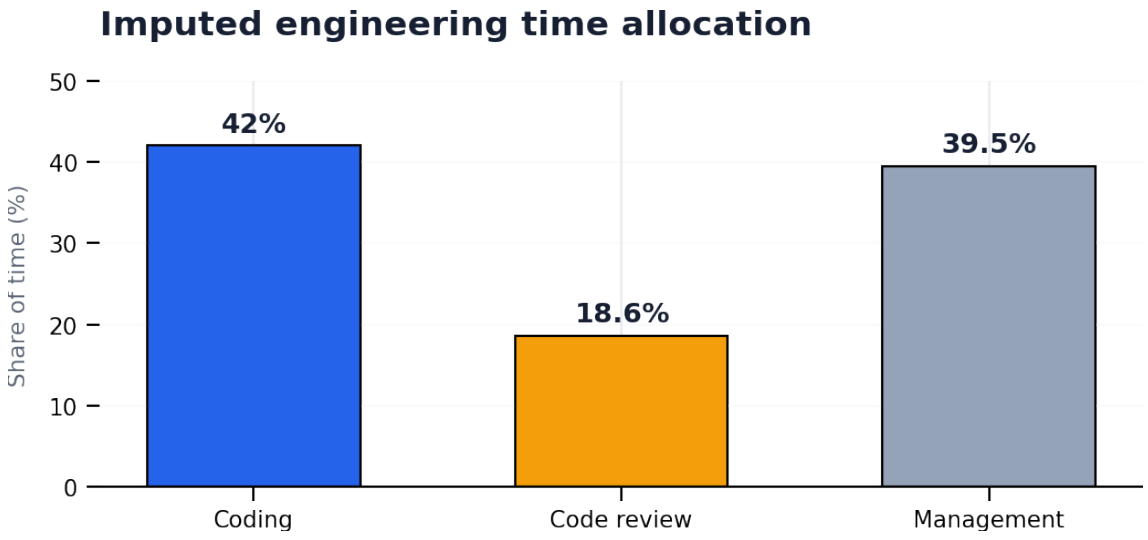
The share of PRs with requested changes rises from a 13% baseline by 12 percentage points.

PR size did not significantly increase, so larger reviews alone do not explain the result.

Source: Chen & Stratton, Artificial Intelligence in the Firm: Bottlenecks in Software Production (Aug. 4, 2026)

WHY IT HAPPENS

Coding is only one part of the production system



Engineering capacity is distributed across coding, review, planning, coordination, testing, and deployment.

Productivity channel

AI increases the volume of draft code.

Quality channel

AI can increase verification required per unit of code.

Organizational response

Firms shift more workers into review.

Source: Chen & Stratton, Artificial Intelligence in the Firm: Bottlenecks in Software Production (Aug. 4, 2026)

THE MECHANISM

The constraint moves downstream



AI expands the queue entering review

The system only moves as fast as its constrained stage.



MANAGEMENT IMPLICATIONS

Optimize the software-production system, not just the coder

1

Measure flow, not activity

Track lead time, review time, change-request rate, escaped defects, deployment frequency and value delivered.

2

Scale verification capacity

Invest in automated tests, CI feedback, security checks, reviewer availability and clear approval ownership.

3

Keep humans at critical gates

AI review adoption is widespread, yet human oversight remains central in the observed review process.

4

Rebalance work deliberately

Expect AI adoption to shift engineering effort toward review, validation and architectural judgment.

Executive takeaway: AI adoption is a workflow redesign challenge, not merely a tooling rollout.

Source: Chen & Stratton, Artificial Intelligence in the Firm: Bottlenecks in Software Production (Aug. 4, 2026)